

Zakhar Pashkin

Senior Computer Vision Engineer | AI Product Engineer | OCR, Segmentation, Detection, VLM/LLM Systems

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SUMMARY

Senior Computer Vision Engineer and AI Product Engineer with 7+ years shipping OCR, segmentation, detection, multimodal search, VLM/LLM workflows, backend services, and production ML systems. Strongest where model quality, product behavior, operator workflow, and launch reliability need one owner: dataset quality, model selection, API delivery, UI surfaces, monitoring, review gates, leak checks, and rollback-ready releases.

CORE STACK

AI products: VLM/LLM workflows, OpenAI APIs, agent automation, Telegram mini apps, Chrome extensions, review gates

Modeling: PyTorch, TensorFlow/Keras, OpenCV, YOLO, MMDetection, CLIP/VLMs, Transformers, ONNX conversion

Production: FastAPI, Tornado, REST/gRPC, Docker, Kubernetes, GitHub Actions, GCP, AWS, Cloud Run, monitoring, CI/CD, MLOps

Computer vision: OCR, segmentation, detection, landmarking, tracking, face texture analysis, neural search, multimodal CV

Edge/mobile: ONNX, TFLite, CoreML, TensorRT, PyTorch Mobile, quantization, mobile inference profiling, Flutter/Android/iOS integration

EXPERIENCE

Senior Computer Vision & AI Engineer

Jun 2022 - Present (4 yrs)

Various Startups - Contract

Senior computer-vision and production-ML engineering across startup, independent, and direct client engagements.

- Shipped OCR, segmentation, detection, landmarking, face-search, neural-search, GPT chatbot, and video-AI systems from model design through FastAPI/Tornado services and cloud deployment.
- Optimized TFLite/CoreML/ONNX edge inference to 50-60 FPS for OCR and segmentation workflows.
- Cut labeling cost by 80% during an auto-annotation rollout used by two client apps.
- Reduced rollback time under 60 seconds across cloud microservices with benchmark checks, human approvals, and release gates.
- Selected client engagements across OCR, segmentation, detection, landmarking, face search, facial analysis, medical AI, health/wellness ML, GPT chatbot, and video-AI systems.
- Lumenis BE, AI Software Consultant, Contract: production-ML support in a medical/aesthetic device context.
- Synoptic Technologies Inc, OCR Engineer, Contract: OCR for social-media image understanding.
- BeagleMind GmbH, Deep Learning Engineer, Contract: facial feature analysis and computer-vision model work.
- Gorilla GmbH, Computer Vision and Deep Learning Developer, Contract: long-running computer-vision/deep-learning client engagement active through 2025.

Top skills: Computer Vision, OCR, Facial Analysis, PyTorch, ONNX, TFLite, CoreML, FastAPI, Agentic Workflows

Senior Computer Vision Engineer

Jun 2022 - Jun 2024 (2 yrs 1 mo)

Wombat Apps LLC / Carb Manage - Contract - Redmond, Washington, United States - Remote

Long-running health and fitness ML engagement for a user-facing nutrition product. Led and developed cloud and mobile ML workflows for food, OCR, and nutrition recognition, connecting model behavior, backend services, mobile inference, and release reliability into production-facing product systems.

- Led development of scalable cloud and mobile ML pipelines for food recognition, OCR-based nutrition extraction, and product-facing nutrition intelligence.
- Built and improved model-to-product workflows across data preparation, model training/evaluation, API integration, mobile deployment, and production support.
- Improved food/OCR recognition quality for users by up to 25% across evaluated product flows.
- Reduced mobile inference latency by up to 40% during mobile deployment optimization.
- Reduced cloud inference/API cost by up to 35% through model/backend optimization across cloud services.
- Increased successful nutrition extraction for users by up to 20% across OCR-based nutrition flows.
- Reduced manual review and correction time by up to 50% through OCR/nutrition extraction improvements.

Skills: Machine Learning, Computer Vision, OCR, Food Recognition, Nutrition Recognition, Python, Production ML, Cloud ML, Mobile ML, Backend Integration, Model Reliability

Computer Vision Engineer

Jun 2019 - Jun 2022

CFT - Akademgorodok, Novosibirsk

- Developed classification, object detection, OCR, segmentation, and document-recognition algorithms for fintech and mobile products.
- Optimized energy-meter OCR and segmentation models for stable 50 to 60 FPS on mid-range smartphones.
- Built masked face anti-spoofing and Cyrillic offline signature-verification research systems with deployment-focused error analysis.
- Improved specialized mobile-GPU model quality with neural architecture search and deployment-focused tuning.
- Implemented CLIP pre-validation and Toloka automation to reduce labeling cost and review load through click-driven segmentation workflows.

Computer Vision Project Mentor

Nov 2019 - Jun 2022

CFT apprenticeship corporate program - Part-time

- Mentored pre-internship engineers on production-style CV projects including mobile energy-meter recognition, crowdsourcing with neural networks, and Cyrillic handwriting OCR.
- Reviewed model choices, dataset quality, error analysis, and deployment constraints so student projects could pass engineering review rather than remain demos.

Machine Learning Project Mentor

Oct 2018 - Oct 2019

School of AI - Remote / San Francisco Bay Area community

- Built and supported ML/DL software for analytics and online education, including Flutter product work and GCP infrastructure.
- Helped secure and operate Google Cloud support for a global AI education platform while coordinating remote contributors and volunteers.

SELECTED PROJECT EVIDENCE

Fast OCR ONNX Inference Server

Three-stage OCR API: line segmentation, word segmentation, CRNN text recognition, Docker packaging, and JSON boxes/text responses.

Multimodal Video Search Platform

Keyframes, ASR/OCR, objects, faces, visual embeddings, transcript embeddings, Qdrant/Postgres retrieval, and quality-agent checks.

Full-Face Wrinkle Segmentation Lab

Cosmetic face-region masks, YOLO segmentation, skeletonized line traces, overlays, CSV outputs, timing events, and visual QA gates.

Dermaself Flutter Skin Analysis App

Flutter/Firebase guided capture app with onboarding, questionnaire intake, offline model runtime, ROI gates, and results screens.

Nutrition Label OCR Mobile Lab

SwiftUI food detection and nutrition-label OCR using ML Kit, crop-assisted text recognition, table parsing, and structured results.

CV Repro Lab Skills

Reproducible CV experiment workflow with benchmark gates, review dashboards, browser validation, and 1,294 public ClawHub downloads across two packages.

RECRUITER EVIDENCE CHECKS

AI product scope: Owns model, backend, workflow, UI, and launch reliability rather than stopping at notebooks or demos.

CV depth: Can explain data, metrics, failure modes, conversion path, and edge/runtime constraints for shipped OCR, segmentation, and detection work.

Evidence quality: Public work links to repos, releases, dated counters, and review gates; non-public CV work is summarized without repo or live-service links.

EDUCATION & CERTIFICATIONS

Siberian Federal University, 2007-2012. Additional professional training includes TensorFlow, Deep Learning and Computer Vision, SQL, JavaScript, Kubernetes on Google Cloud, and PyTorch scholarship work.